

TYPES OF BOATS

Boats are organized by class, which is determined by their length. Small boats are ideal for short-distance racing, as they move quickly. Large boats are better for long-distance sailing as they can endure more treacherous seas than a small dinghy. Here are some popular classes of boat.



LASER CLASS

A popular 4.2 m (14 ft) dinghy for solo sailing.



470 CLASS

A 4.7 m (15.4 ft) dinghy for a crew of two.



49ER CLASS

A 4.9 m (16 ft) dinghy with CCA spinnaker for speed.



TORNADO CLASS

A 6.1 m (20 ft) catamaran with two body sections that increase the boat's speed.



OCEAN RACER (VOLVO 70) CLASS

A 21.3 m (70 ft) yacht with a 31.5 m (103 ft) mast.



AMERICA'S CUP CLASS

A 24 m (79 ft) yacht used in the America's Cup race between 1992 and 2007.

SAILING FOR SPORT, RATHER THAN TRANSPORTATION OR WARFARE, BEGAN IN THE NETHERLANDS IN THE 1600s

Spinnaker

Foresail

Port side

Bow

Starboard side

USING THE WIND

Sailors can adjust their sails to make the most of the wind. The sails can be angled to capture the wind, so the boat is pushed forwards in the direction it faces. A sailing boat can travel in any direction except straight into the wind (the no-sail zone).

KEY

No-sail zone

In this area the boat would be sailing into the wind. The sails would not work and the boat would not move.

Close haul

The closest a boat can sail to the wind without entering the no-sail zone. Both sails are pulled in tight to the centreline.

Close reach

Similar to a close-hauled course, but the boat is turned away a little more from the wind and the sails are loosened further.

Beam reach

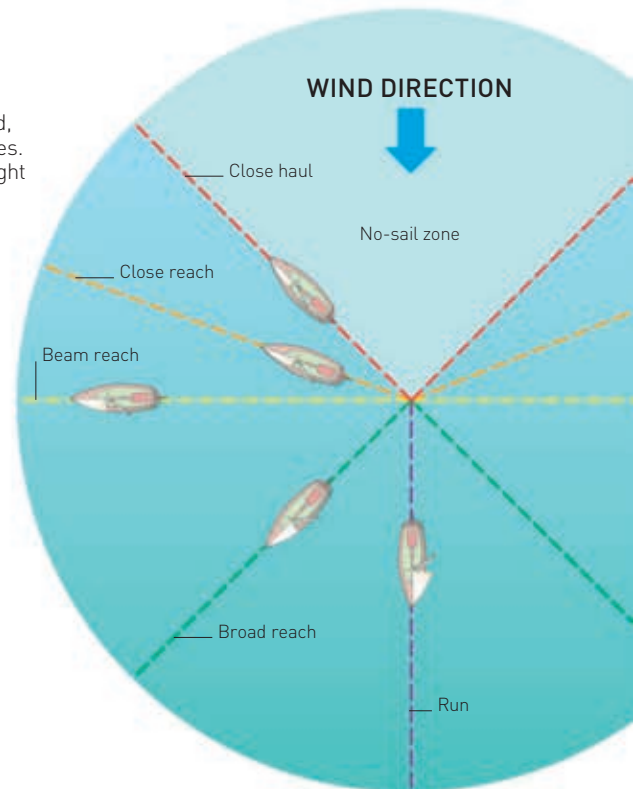
Sails are eased halfway and the wind is coming directly across the side of the boat.

Broad reach

Sails are nearly full and the boat is on a course away from the wind (downwind).

Run

Sails are full and the wind is directly behind the boat.



TACKING AND GYBING

There are two ways of turning a boat: tacking and gybing. Tacking is a safer, slower way of turning as it allows more control of the sails. Gybing is faster and is especially good for racing.



TACKING

Turn the boat to face upwind.



GYBING

Turn the boat to face downwind.

RECORD BREAKERS

Since sailing began as a sport several hundred years ago, many sailors have set impressive around-the-world sailing records.

- **JOSHUA SLOCUM (CANADA), 1895–98**
The first person to sail solo around the world, with just three stops.
- **ROBIN KNOX-JOHNSTON (UK), 1969**
The first person to sail solo around the world without stopping.
- **KAY COTTEE (AUSTRALIA), 1988**
The first woman to sail solo around the world without stopping.
- **ELLEN MACARTHUR (UK), 2005**
Became the fastest person to sail solo around the world without stopping, in 71 days, 14 hours, 18 minutes, and 33 seconds.
- **FRANCIS JOYON (FRANCE), 2008**
Broke Ellen MacArthur's record to become the fastest person to sail solo around the world without stopping, in 57 days, 13 hours, 34 minutes, and 6 seconds.

OCEAN RACING

Ocean races can be extremely challenging and dangerous. They require both physical and mental strength, as sailors can be at sea for many weeks at a time.

ROUTE DU RHUM

Singlehanded racers must work with fast winds in this high-speed journey across the Atlantic Ocean.

VOLVO OCEAN RACE

In this extreme race, nine-person crews sail around the world day and night.

KEY

- Route du Rhum
- Volvo Ocean Race
- Velux 5 Oceans
- South Atlantic Race



VELUX 5 OCEANS

This ambitious solo round-the-world race takes more than 100 days to finish.

SOUTH ATLANTIC RACE

Teams in this race must face the strong winds and huge waves of the southern Atlantic Ocean.